Effectivity

All LightSpeed QX/I Systems manufactured befoe August 11, 2000 with Gantry Model Number 2180482. Some of the systems near this manufactuing cut in date may have the new revision of Software loaded. If the system has LightSpeed QX/I Applications Software Version 1.2 M4 (run "showprods" to determine software version), the FMI is not required. Otherwise, the FMI is required.

Purpose

This FMI is the method of distribution being used to distribute the upgrades and bug fixes provided by this new release of software. The features introduced with this new software include (but are not limited to) the following:

- Low Signal Performance (LSP)
- Single Slice (1 X 1.25)
- Multi Slice Prospective Multi-Recon (MS-PMR)
- Image Analysis Tool
- Web Browser Based Common Service Desktop
- Common Software with forward production systems built without O2 computers
- Patient Procedure Step (PPS requires Connect Pro Option)
- Direct 3D (a purchased Option)
- Prospective Gating (a purchased Option)
- CT Perfusion on Console (a purchased Option)

Prerequsite FMI's

FMI 25261 LightSpeed 1.1 Z Axis Tracking

Related FMI's

FMI 25266 LightSpeed 1.2 Translated Operator Manuals.

Furnished Materials

FMI Kit Part Number 2276703 includes the items listed below

Qty	Description	Part #
1	FMI Document	2274110-100
1	LightSpeed QX/i System Software CDROMs	2260334-3
1	LightSpeed QX/i System Service Manual General	2278883-100
1	LightSpeed QX/i System Installation Manual General	2278881-100
1	LightSpeed QX/i Illustrated Parts Manual	2211225-100
1	LightSpeed QX/i Schematics and Diagrams Manual	2211227-100
1	LightSpeed QX/i Functional Interconnects Manual Advanced	2211228-100
1	LightSpeed QX/i Technical Reference Manual	2211214-100
1	LightSpeed QX/i Load From Cold Manual	2211220-100
1	LightSpeed QX/i Application Tip Tutorial CDROM	2226999
1	LightSpeed 1.2 Service Introduction Planner	2275682-100
1	LightSpeed QX/i Applications Tips and Work Arounds	2211223-100
1	IC Removal/Insertion tool	2276193
1	Recon Interface Processor (RIP) Firmware IC XU1	2276195
1	Recon Interface Processor (RIP) Firmware IC XU2	2276197
1	Antistatic Wrist Strap	2252691

Tools Required

MOD for Save State

Standard Field Engineer Tool Kit (including antistatic mat/kit)

Poly and QA Phantoms that are on each site

InSite Package (Distributed on Tab 182)

- InSite Software (GE Part Number 2243504-5)
- InSite Installation Manual (Direction 2211226-100)

People Power

Labor Estimates:	6 hours
Travel Estimates:	1 hour

Procedure

Introduction

This FMI includes the following activities:

- Preparation for Upgrade
- RIP Motorola Firmware Upgrade
- Load From Cold (LFC)
- System Calibrations
- IQ Checks
- InSite Reconfiguration
- Save System State
- System Checkout
- FMI Completion

Install Sequence Guide

Refer to Figure 1 for a LightSpeed 1.2 Upgrade Installation Sequence Guide.

Figure 1

Light Speed 1.2 Software Release Upgrade Installation Sequence Guide

Prepare System For Upgrade

- ❑ Software Revision Check
- ❑ Archive Patient Images
- ❑ System Operation Verification
- ❑ Remove DASM from SCSI Chain (If appliable)
- ❑ CDROM Drive Verification
- ❑ System Data Archive (Save System State)
- ❑ Record Key System Information

FMI 25263 Section 1
Direction 2211220-100

Upgrade RIP Board Firmware

- ❑ Recon Interface Processor (RIP) Firmware Upgrade (only if required)
- ❑ Re-Seat Scan Data Corrections (SDC) Board

FMI 25263 Section 2

Load From Cold Procedure (Initial)

- ❑ Shutdown the System
- ❑ Boot Standalone IRIX Disk Partioning Utility
- ❑ Install System OS Software on the Operator Console
- ❑ Install System Software on the SBC
- ❑ Install QX/i Software on the OC and SBC

FMI 25263 Section 3
Direction 2211220-100

CAUTION

AVOID LOSING CAL DATA!
On the Hardware Settings Screen of the Reconfig Menu, ensure that the "`IS O2 PRESENT`" selection is set to "`YES`"!! Setting this selection to "`NO`" will cause the system to crash and force another LFC.

Load From Cold Procedure (Finish)

- ❑ Set Date/Time on SBC & OC
- ❑ Start up CT Applications
- ❑ Restore System State
- ❑ Downloading Flash Firmware
- ❑ Add Software Options

FMI 25263 Section 4
Direction 2211220-100

NOTE: Be sure to enable all previously enabled LIghtSpeed 1.1 Release options using the on-site option MODs.

Calibrate 1x1.25 Single Slice

- ❑ DAS Gain Calibration
- ❑ Collimator Calibration
- ❑ Detailed Calibration (1x1.25 Slice Only)
  - ❑ Select kV's: **80, 100, 120, 140 kV** (as applicable)
  - ❑ Select appropriate SFOV's (Small/Head, Large/Body)
  - ❑ Select new slice collimation: **1x1.25**
  - ❑ Select appropriate Focal Spots (Small, Large)
- ❑ Auto CT # Adjust
- ❑ Perform Image Quality Checks

FMI 25263 Section 5
Direction 2278881-100

Reconfigure InSite

- ❑ Class M Install
- ❑ InSite Interactive Configuration
- ❑ InSite Auto Checkout

FMI 25263 Section 6
Direction 2211226-100

Complete Upgrade

- ❑ Save System State
- ❑ System Checkout
- ❑ FMI Completion

Direction 2211220-100

Section 1 Prepare System For Upgrade

Software Revision Check

The software revision can be checked using the showprods command.

1.) Open a UNIX shell.

2.) Type:

`showprods | grep -i "QXi Apps" <ENTER>`

3.) Review the response to the above command against the three scenerios listed below. It is critical that the software be at LightSpeed 1.1M3 or higher prior to completing this FMI.

`I.    LightSpeed 1.0 M4 Release:`

Helios_OC_Product 05/07/1999 QXi Apps ver 4.1_2.7.3gpc2

Stop!

Perform FMI 25261 (Includes Hardware, Software, Calibrations required to support the 1.2 Upgrade).

`II.   LightSpeed 1.1 M4 Release:`

Helios_OC_Product XX/XX/XXXX QXi Apps ver 6.8.2_2.7.3n2

Continue with FMI 25263.

`III.  LightSpeed 1.2 M4 Release:`

Helios_OC_Product XX/XX/2000 QXi Apps Ver 8.X.X_2.8.1.X_HE1.2M4

FMI 25263 in not needed.

Image Data Archive

The following procedures will destroy all image data on the system. Verify that the customer has reconstructed and archived all images before proceeding.

System Operation Verification

Setup and perform at least one axial scan to verify that the system operated properly before beginning the upgrade.

Remove DASM From SCSI Chain

During the load of applications software, the SCSI bus is being heavily utilized. During this process there have been sites that have experienced load problems. To minimize the occurrence of these problems, it is recommended to remove the DASM from the SCSI bus chain before continuing with this FMI. Once the LFC has been performed the DASM can be reinstalled.

1.) At the SCSI Expansion box disconnect the cable coming from the DASM.

2.) At the DASM, remove the SCSI cable that comes from the Octane and conect it to the SCSI Expansion box connection opened in step one.

CDROM Drive Verification

Since the CDROM Drive is seldom used except for loading system software, it is advisable at this time to perform a quick test on the ability of the CDROM drive to read data from a CD before you begin the LFC procedure. This test is accomplished by inserting and mounting one of the OS CDROM disks, and performing a checksum (read check) on the files in one of the directories on the CD. If any errors occur while the system is reading this CDROM, you must stop to investigate and correct the CD problem (for example the SCSI Bus, CDROM Drive, etc. problems).

1.) Open a UNIX shell.

2.) Type:

`tail -f /usr/adm/SYSLOG <ENTER>`

3.) Open a second UNIX shell.

4.) Insert the LightSpeed QX/i IRIX Operating System CDROM Disk #1 into the CDROM Drive and type:

`mountCD /mnt <ENTER>`

`cd /mnt/dist <ENTER>`

`sum * > /tmp/CDROMtest <ENTER>` *(~5 minutes to display prompt)*

5.) While the system is reading the CDROM and calculating checksums, you need to monitor the SYSLOG in the first UNIX shell window for any error messages that may occur that are related to SCSI or CDROM problems (timeouts, read errors, SCSI Bus resets, etc.).

6.) If checks are successful (no errors produced), type the following command to verify the checksum:

`sum /tmp/CDROMtest <ENTER>`

7.) Verify that the resultant display for the command in step 6 is:

34499 8 /tmp/CDROMtest

8.) If you encounter any CDROM or SCSI related errors during this check, stop and troubleshoot/ repair the problem before attempting to perform the FMI.

9.) Terminate the 'tail' command by typing `^c` in the first UNIX shell window where the 'tail' command is running.

10.) Unmount the CDROM by typing the following commands in the second UNIX shell window:

`cd / <ENTER>`

`unmountCD <ENTER>`

System Data Archive (Save System State)

This section details the procedure for archiving the system critical files (System State) and adding to them the customer protocols that previously were not included in the System State archive process. This newly created System State will be used during the upgrade Load From Cold Procedure. In the future, 1.2 software will archive the protocols as a part of the System State archive.

1. Create a System State MOD (refer to the LightSpeed Load From Cold Manual, section 2.4 for instructions).
2. With the System State MOD still in the drive, mount the drive using the command:

```
mountMOD <ENTER>
```

3. Verify that the directory structure (service_mod_data/system_state) exists by typing:

```
cd /MOD/service_mod_data <ENTER>
ls <ENTER>
```

NOTICE: If the directory structure does not exist repeat step 1 above (create a System State MOD).

4. Copy the Voxtool Prefs files to the System State MOD by typing the command:

```
cp /usr/g/ctuser/Prefs/VoxtoolPrefs/*.* /MOD/service_mod_data/system_state/
<ENTER>
```

NOTICE: If the message "No such file or directory" appears, it means you have not typed the string correctly or there are no files in the VoxtoolPrefs directory to archive. If your typing was correct, move to step 5. If not, retype the command and try again.

5. Verify that all files (if any existed) have been saved by typing the command:

```
ls -l /MOD/service_mod_data/system_state/ <ENTER>
```

6. The unmount command can not be run while in the MOD directory. To perform the unmount type the commands:

```
cd <ENTER>
unmountMOD <ENTER>
```

Section 2 Upgrade RIP Board Firmware

Recon Interface Processor (RIP) Firmware Upgrade

With the 1.2 release of system software, the RIP flash firmware will be flashed from the system software. This function will only work on systems with PPC1/3.6 or higher firmware. A set of Firmware ICs have been provided with this FMI. It is however recommended that you not change them if you do not need to.

CAUTION: THE IC SOCKET CAN BE DAMAGED DURING IC REPLACEMENT!

To check the current firmware version on the RIP board the RIP board Jumper J15 must be placed in position 2-3.The following steps detail the process for moving Jumper J15 to the correct position.

Static Sensitive **Use Static mat and wrist strap to prevent circuitry damage!**

1. Shutdown the software and turn off the power to the console.

2. Open the front of the console and remove the cover from the EMC side.
3. Remove the board hold-down strap from the RIP board.
4. Remove the RIP circuit board from the Host Console.
5. Move the Jumper J15 to position 2-3 (see figure 1 for location).
7. Install the RIP circuit board back into the console assuring that it is securely seated in place.
7. Stop the system from booting to applications.
8. Open a Unix shell.
9. log into the SBC:

   rsh sbc <Enter>;

To get the revision information:

10. Connect to the card

    cu ice <Enter>;

11. Place the card in PPCBUG (lowest boot level) mode

    A - press **<Enter>**; if PPC1-Bug> is the prompt, skip to step 13

    B - If not at PPC1-Bug> prompt, press the reset button on the front of the card

12. Verify that the card is in PPCBUG (see step 11)
13. Get the revision information

    ver

A session showing this is as follows...

```
% cu ice

PPC1-Bug>

PPC1-Bug>ver

Debugger/Diagnostics

Type/Revision............=PPC1/3.6 (old version is 3.5)

Debugger/Diagnostics Revision

....................................

....................................
```

14. In the following step you wil use the ~ (tilde) key. The keystrokes to aquire this selection are different dependant on the language of your keyboard (example - The standard english key board requires the shift key and the ~ ` key be depressed).

    Return to the Octane ctuser promt by typing:

    <~> <.>

    Within 10 seconds press: **<Enter>**

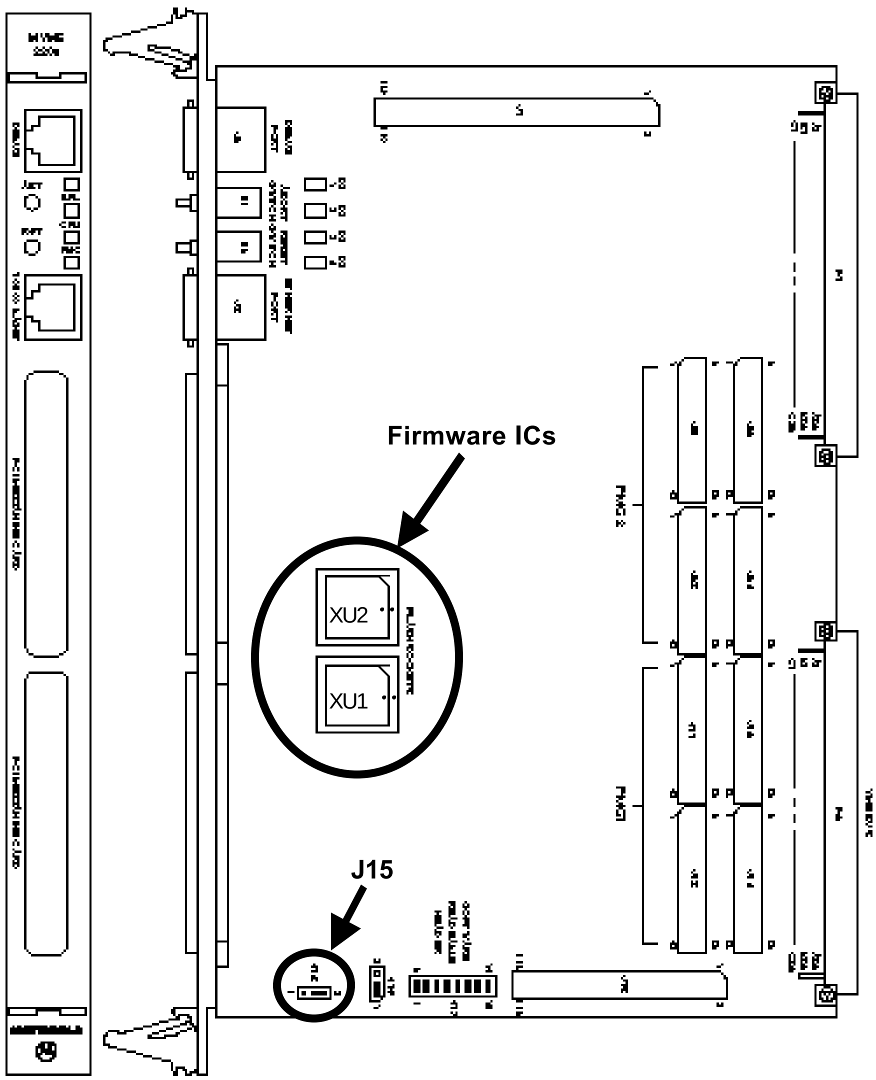
If your RIP firmware is at a version 3.6 or higher, skip to Software Load Procedure.

If your RIP firmware is at a version below 3.6 you will need to replace the Ics Xu1 and Xu2 on the RIP board before performing the software load. The following instructions detail the replacment of these firmware IC.

Static Sensitive **Use Static mat and wrist strap to prevent circuitry damage!**

1. Shutdown Software and turn off the power to the console.

2. Remove the RIP circuit board from the Host Console (see Direction 2211221-100 Chapter 4 Console Physical Organization to locate RIP board)
3. Move RIP board to the static safe work surface for IC removal.



Recon Interface Processor

Figure 1

CAUTION: THE IC SOCKET CAN BE DAMAGED DURING IC REPLACEMENT! LET THE TOOL PULL THE CHIP FROM THE SOCKET!

4. Using the supplied IC removal tool, remove and set asside XU1 and XU2 from the RIP board (see Figure 1).
5. Install the supplied Revision 4.1 XU1 and XU2 taking care to align the cut corner of the IC with the cut corner of the IC socket. An additional alignment dot is on both the IC and Socket.
6. Verify Jumper J15 is in position 2-3 position (see Figure 1). This enables FLASH capability.
7. Mark the board faceplate with the part number 2197234-4
8. Install RIP circuit board back into the Host Console assuring that it is securely seated in place.
9. Reinstall the board hold-down strap removed in step 3 above.
10. Reinstall EMC cover.
11. Leave front cover off until LFC and Calibrations have been completed.

Re-Seat Scan Data Corrections (SDC) Board

NOTICE: Removing and re-installing the RIP circiut board in the VME Chassis may cause the SDC Mercury circuit board to become unseated or partially seated in the VME Chassis backplane. To ensure the SDC Mercury circuit board is properly seated, we recommend that you re-seat the SDC Mercury circuit board any time you remove/install the RIP board.

Section 3 Load From Cold Procedure (Initial)

After system operation has been verified, archive has been completed, and the RIP firmware changed, you are ready to upgrade the system software.

Using the LightSpeed QX/i Load From Cold Manual (2211220-100) and the software included in the FMI kit, perform a software load. **Refer to Section 2.6 through 2.11 of the LFC manual**, follow the procedure to load the software and restore the System State from MOD.

CAUTION: AVOID LOSING CAL DATA!!

On the Hardware Settings Screen of the RECONFIG menu, ensure that the IS O2 PRESENT **selection is set to YES!! Setting this selection to NO will cause a system using an O2 Computer to crash. This will force you to perform another Load From Cold.**

Section 4 Load From Cold Procedure (Finish)

CAUTION: FLASH FIRMWARE UPDATES REQUIRED

Flash download is required and will take 10-15 minutes. During this process there will be error messages indicating lack of communication. Please be patient while this process takes place. The "stop" key will go inactive when download is finished.

Using the LightSpeed QX/i Load From Cold Manual (2211220-100), complete the Load From Cold procedure as described in **Sections 2.12 through 2.16 of the LFC manual**.

Section 5 Calibrate 1x1.25 Single Slice

System Calibraton

Due to the addition of Single Slice (1 X 1.25mm), it is necessary to run a full set of air calibrations and phantom calibrations for this mode. The recommendation from the engineering team is to step through the full set of calibrations and assure that each calibration impacted by 1 X 1.25 is performed. This includes **DAS Gain Cal., Collimator Cal., Detailed Cal**. and **Auto CT # Adjust**. The longest this calibration should take will be 4.0 hours. For instructions see LightSpeed Installation Manual (2278881-100or 2278883-100).

IQ Checks

Using the information provided in the System Installation Manual (2278881-100), Chapter 10 System Tests, perform the QA Image checks.

Section 6 Reconfigure InSite

Following the instructions in the InSite Installation Manual (Direction 2211226-100), load the InSite software and perform auto checkout.

Section 7 Complete Upgrade

Save System State

Refer to the LFC Manual (2211220-100), and perform a Save System State. This will save the new calibrations as well as the customer protocols.

System Checkout

Refer to the LFC Manual (2211220-100), and perform the system performance checks as described in the manual. Ensure all networking and filming processes are still functional.

Once system checkout has been performed, you can install the front cover of the console.

FMI Completion

Complete the Product Locator information cards and return.

Fill out and return the FMI Completion Certification Report paperwork (or electronically close your FMI dispatch) in accordance with your local procedure.

Addendum

Manual Flash RIP Firmware

During the LFC, the system performs a firmware flash to the RIP board. If the flash does not work properly, the system will attempt to boot to applications level and will not. If this happens, you may need to identify the cause of the fault and perform a manual pflash. To perform the firmware flash follow the steps below.

1. Shutdown the applications software:

   Service Desktop

   └▶ ***Tools/Utilities***

   └▶ ***Applications Shutdown***

   Wait 30 to 60 seconds for applications to shutdown before going to step 2

2. Open a Unix shell

3. Log into the SBCusing the command:

   `rsh sbc <Enter>`

4. Flash the firmware. Using the command:

   `/usr/g/ice/bin/pflash <Enter>`

If flash works properly, you will see loading type text in approximatly 12 lines. If the process fails, you will get an error message at which point you will need to verify the jumper position of J15 and the firmware revision (see steps RIP Firmware Upgrade section).

5. Configure the Scan Data Disk using the command:

   `/usr/g/scripts/reconfigScanDisk <Enter>`

6. At this point you will need to return to seciton 4 and continue the upgrade process.